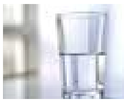
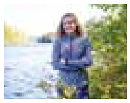


Combating biasis

Humans have been wired to delude themselves into taking counterproductive decisions

https://dgit.in/sep18-71-1



How a teen saved a river

A determined teenager, scientific tests, and a Facebook page saved this contaminated river

https://dgit.in/sep18-71-2

the Planet

possibility that the rainfall in Asia, South America and Africa would be disrupted by the aerosols. Ultimately, the effects are not fully understood, and they may not be as beneficial as they were originally thought.

An agricultural economist at UC Berkeley, Jonathan Proctor studied the possible effects of aerosol SRM techniques on crop yields. Environmentalists had assumed that crop yields would also increase because of the effect seen on plants and trees after the Pinatubo eruption. As the crops were spaced apart, as against tightly packed together in a jungle, there was no discernable difference for most crops. The yields for some crops, such as corn, actually decreased.

There are many other proposed ideas for increasing the reflectivity of the planet, from filling the oceans with reflective foam or plastic islands, to painting mountains white. One of the ideas that sounds like it is straight from science fiction, is building a large mirror, or a swarm of mirrors at the L1 gravitational point between the Sun and the Earth. At that point, the mirror would permanently shield the Earth from a portion of the sunlight. The mirror would have to have an area of 1,600,000 square kilometers, to deflect just one percent of the sunlight. An alternative is to place a sunshade at the same point. Another idea is to put trillions of 2 feet wide, ultra thin lenses that each weigh less than a butterfly into orbit. The purpose is to deflect sunlight away from the Earth. The effects of such reflectors and shades are not fully understood, and could lead to changes in the seasons, or shifts in the ocean currents.

One of the major problems with SRM approaches is that while the amount of sunlight heating the planet is reduced, it does not really change the build up of carbon dioxide within the atmosphere. Global warming is just one effect of indiscriminate dumping of carbon dioxide gases in the atmosphere. Another effect is the acidification of the oceans, because the oceans and water bodies absorb between 30 to 40% of the carbon dumped in the atmosphere. Even if the SRM approaches do work, the oceans and marine life for example, would continue to be affected.

CARBON DIOXIDE REMOVAL

The other significant approach is to find techniques to remove the carbon dioxide we have already pumped into the atmosphere. Over the course of the next decade, many countries in the world are expected to contribute by undertaking carbon sequestration programs – using various approaches to capture and store carbon dioxide in the atmosphere. The tried and tested - and natural - way of doing this is planting tons of trees. However, that would require significant investment in terms of land, which is currently being put to agricultural use.

Klaus Lackner, an engineer at Arizona State University has developed an artificial tree that is a thousand times better than natural trees at sucking carbon dioxide out of the atmosphere. The trees lining highways and public spaces could contribute to the removal of carbon dioxide from the atmosphere. We would require one hundred million of Lackner's trees, to remove the amount of carbon dioxide dumped into the atmosphere every year. There are technologies that can be used to convert the captured carbon dioxide back into fuel, providing a new source of energy.

Another interesting possibility is a process known as ocean fertilization. The idea is to put iron, urea, nitrogen or phosphorus into the oceans. This will increase the amount of marine food production by algae blooms. The blooms will in turn absorb carbon dioxide from the atmosphere, and sink to the sea floor with it after they die. These nutrients are not normally available in the oceans in the quantities required, which is why they have to be artificially introduced. Iron has proven to be particularly effective, and over 12 real world experiments have been conducted since 1993. 83,000 kg of carbon can be removed from the atmosphere with just 1 kg of iron. The absorption of iron is particularly effective in the cooler regions, so the closer to the pole it is dumped the better. However, ocean fertilisation is also not without its own problems. The large blooms of algae might consume the carbon dioxide, but they also use up significant quantities of oxygen at the same time. This means that the conditions are made challenging for other life forms in the seas.

We have been at a critical juncture in the human civilisation for some time now. According to the The Intergovernmental Panel on Climate Change, global temperatures may rise by as much as 5.8°C by the end of 2100. A 2° change in global temperatures is considered too much by scientists and environmental activists, and it would result in a significantly different planet. The Paris Agreement aims to keep global temperatures from rising this much, as compared to pre industrial levels. The scientists are researching the possibilities of geoengineering, despite knowing that the ideas are outlandish, unfeasible, impractical, and even irresponsible. That should be some indication of exactly how dire the situation is.